

Lab 1

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1 Introduction

Even though electronic health records (EHRs) are now commonly used throughout healthcare, sharing patient information between different healthcare organizations is still a challenge. Hospitals, primary care providers, specialists, and other organizations may use different systems that do not always communicate well with each other. Because of this, staff may have to rely on methods such as faxing records, making phone calls, or manually following up on referrals. When records are missing or a provider cannot easily see the status of a referral, the process can take longer and create additional work for healthcare staff. More importantly, these delays can affect how quickly a patient receives the care they need. According to the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (ASTP/ONC, 2024), 70% of hospitals participated in all four areas of interoperable health information exchange at least sometimes, but only 43% did so routinely. This shows that while healthcare technology has improved, there are still gaps in how consistently patient information is exchanged.

A solution to this problem needs to make communication between healthcare organizations easier without expecting them to completely replace the systems they already use. Providers need a way to securely send the information needed for a referral, see where that referral is in the process, and know when something requires their attention. At the same time, patients should have more visibility and control over who has access to their information. Since medical records contain sensitive information, security also has to be an important part of the solution. Features such as authentication, access controls, consent management, and activity tracking are needed to make sure patient information is handled appropriately.

PatientPath is our proposed solution to this problem. PatientPath is a web-based platform that helps healthcare organizations share medical records and manage patient referrals in one place. Instead of replacing an organization's existing EHR, PatientPath is meant to work alongside it and help connect organizations that may otherwise use separate systems. Providers can create referrals, choose which patient records need to be shared, track the progress of a referral, and see when a referral may need additional attention. Patients can also manage which organizations or providers have permission to access their information. PatientPath will use healthcare interoperability standards such as FHIR and HL7 to support the exchange of information between different systems.

For CS 410, our prototype will focus on demonstrating the main features of PatientPath rather than building a complete healthcare system. The prototype will show how a provider can create a referral, select records to share, and follow the referral as its status changes. It will also demonstrate patient consent controls, activity history, and tools for identifying referrals that have been waiting too long. Referral statuses such as Pending, In Review, Accepted, and Complete will make it easier to see where a referral is in the process. By demonstrating these features, the prototype will show how PatientPath could make the referral process more organized, transparent, and efficient for both healthcare providers and patients.

PatientPath is a web-based platform designed to make sharing medical records and managing patient referrals easier between healthcare organizations. The platform is not intended to replace the EHR systems that hospitals and providers already use. Instead, it works as a connection between those systems by giving providers one place to manage the information and communication involved in a referral. A provider can create a referral, choose the patient records that need to be shared, send the necessary information to another provider, and track the referral as it moves through the process. Statuses such as Pending, In Review, Accepted, and Complete allow providers to quickly understand what is happening with each referral. PatientPath will also identify referrals that have been inactive for too long so staff can follow up before they are overlooked. Patients will have their own consent controls where they can see which organizations have access to their information and grant or revoke that access when needed. Overall, the goal of PatientPath is to reduce the amount of manual communication involved in referrals while making the process easier to follow. By improving how providers share information and track referrals, PatientPath is intended to reduce delays, keep patient information secure, and help patients move between healthcare providers more smoothly.

2 Product Description

PatientPath is a web-based platform designed to improve how healthcare organizations share patient information and manage referrals. The system provides a centralized place where primary care providers, specialists, and other healthcare staff can create referrals, select the necessary patient records to share, and track a referral throughout the process. Referral statuses such as Pending, In Review, Accepted, and Complete allow providers to quickly see where a patient is in the referral process and identify when additional follow-up may be needed.

PatientPath is designed to work alongside existing electronic health record (EHR) systems rather than replace them. Healthcare organizations often use different systems that do not communicate easily with one another, which can lead to phone calls, faxes, repeated requests for records, and delays in patient care. PatientPath addresses this by using healthcare interoperability standards such as FHIR and HL7 to support the exchange of information between different systems. This allows healthcare organizations to continue using their existing EHRs while using PatientPath as a connection between them.

Patients are also included in the process through the Patient Portal. Patients can view information related to their referrals and manage consent for access to their information. Security features such as authentication, role-based access control, consent management, and activity tracking help protect patient information and control how it is accessed. By bringing referral tracking, record sharing, patient consent, and communication into one platform, PatientPath is intended to create a more organized and transparent referral process for both healthcare providers and patients.

2.1 Key Product Features and Capabilities

PatientPath brings medical record sharing and referral management together in one platform. Providers can create a referral, select the patient information that is needed, securely send it to another provider, and track the referral throughout the process. Each referral is given a status, such as Pending, In Review, Accepted, or Complete, so providers can quickly see what stage it has reached. The system can also identify referrals that have been inactive for a certain amount of time, helping staff recognize when follow-up may be needed. Patients have access to consent controls that allow them to see which providers or organizations have access to their information and grant or revoke that access.

One of the main features that makes PatientPath different is that it is designed to work alongside existing electronic health record systems rather than replace them. Healthcare organizations have already invested in their own EHR systems, so requiring everyone to use the same system would not be realistic. PatientPath instead acts as a connection between organizations by using healthcare interoperability standards such as FHIR and HL7. This allows information from different systems to be shared through a common platform while still allowing each organization to continue using its current EHR. PatientPath also combines record sharing, referral tracking, and patient consent into one process instead of treating them as completely separate tasks.

The overall purpose of PatientPath is to make referrals easier to manage and less dependent on manual communication. When providers have a clear view of a referral's status and the necessary records are available in the same system, there is less need for repeated phone calls, faxes, or follow-up requests. Alerts for referrals that have been waiting too long can also help prevent patients from being overlooked during the referral process. By improving communication between providers while giving patients more control over their information, PatientPath addresses both the technical and organizational problems that can cause delays when patients move between healthcare organizations.

2.2 Major Components (Hardware/Software)

PatientPath is mainly a software-based product and does not require any specialized hardware for the prototype. Providers, specialists, administrators, and patients can access the system using a computer, laptop, tablet, or other device with an internet connection and web browser. On the system side, PatientPath requires a server environment to run the application and a database to store information. Since the purpose of the prototype is to demonstrate the functionality of PatientPath, features that would be required for a full production environment, such as multi-cloud hosting, high-availability infrastructure, and a cloud-based key management system, are outside of the prototype's scope.

As shown in the Major Functional Component Diagram (MFCD), PatientPath is divided into three main layers: the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer is what users directly interact with and includes the Provider Dashboard, Patient Portal, Admin Console, Login and MFA Screens, and Notification Interface. The Application Layer handles the main functions behind those interfaces. This includes Secure

Record Exchange, Identity and Access Management, Patient Record Management, FHIR/HL7 Integration, and the API Gateway. Finally, the Data Layer manages the information used by PatientPath through Encrypted Document Storage, Audit Logs, Consent Records, FHIR Normalized Data, and a PostgreSQL Database. Separating the system into these layers helps keep the user interface, application functions, and stored information organized while allowing the components to work together.

The software developed for the PatientPath prototype will focus on demonstrating the most important parts of the referral and record-sharing process. The Provider Dashboard will allow PCPs and specialists to manage referrals, while the Patient Portal will allow patients to view records, referral statuses, and consent information. The Admin Console will support referral monitoring, auditing, and role-based access control. Behind these interfaces, PatientPath will support secure record sharing, patient record management, access permissions, consent records, audit logs, and real-time referral status updates. Some functionality, including MFA, FHIR/HL7 integration, the API Gateway, encrypted document storage, and FHIR-normalized data, will only be partially implemented in the prototype. Direct EHR integrations, external FHIR connections, legacy HL7 messaging, multi-cloud hosting, high-availability infrastructure, and cloud key management will remain outside of the CS 410 prototype scope. This allows the prototype to demonstrate how PatientPath would work without requiring the resources and access needed to build a full production healthcare system.

3 Identification of Case Study

PatientPath is being developed for healthcare providers and patients who experience the challenges that come with sharing medical information between different healthcare organizations. Primary care providers (PCPs), specialists, and administrative staff need a more organized way to send referrals, share the necessary patient records, and keep track of where a referral is in the process. Patients are also an important part of PatientPath because the platform gives them more visibility into their referrals and control over which providers or organizations can access their information.

For the prototype, the case study group will consist of a small group of users representing the four main roles within PatientPath: PCPs, specialists, administrators, and patients. These users will interact with the prototype from their assigned perspective and provide feedback on the system. For example, providers can test whether creating and tracking a referral is easy to understand, while patients can provide feedback on the consent and referral information available to them. This feedback can help identify areas that may need to be changed or improved as PatientPath continues to develop.

In the future, PatientPath could be used by a much larger range of healthcare organizations and professionals. This could include hospitals, private practices, specialty clinics, Federally Qualified Health Centers (FQHCs), and other organizations that regularly send or receive patient referrals. Additional staff involved in medical records, referral coordination, compliance, and healthcare administration could also benefit from the platform. Although these users may have different responsibilities, PatientPath is intended to give them a more connected way to share information and follow a patient throughout the referral process.

4 Product Prototype Description

The PatientPath prototype will demonstrate the main functionality of the real-world product developed during CS 410. The goal of PatientPath is to create a more connected referral process between primary care providers, specialists, patients, and administrators. The prototype will focus on showing how a referral moves through the system from beginning to end. This includes creating a referral, selecting patient records to share, tracking the referral's status, managing patient consent, and identifying referrals that may require follow-up.

Because the prototype is being developed within the scope of CS 411W, it will not include every capability expected in the real-world version of PatientPath. Some features will be reduced, simulated, or modeled rather than fully implemented. The prototype will not connect directly to real electronic health record (EHR) systems or use actual patient health information. Direct EHR integrations, external FHIR connections, legacy HL7 messaging, multi-cloud hosting, high-availability infrastructure, and cloud-based key management will remain outside the scope of the prototype. Other features, including MFA, FHIR/HL7 integration, the API Gateway, encrypted document storage, and FHIR-normalized data, will only be partially implemented.

Reducing or simulating these features allows the team to focus on the main PatientPath workflow while still demonstrating how the complete system would function in a real healthcare environment. The prototype will act as a proof of concept for how PatientPath can improve communication, referral tracking, and patient involvement without requiring access to real healthcare systems or sensitive patient information.

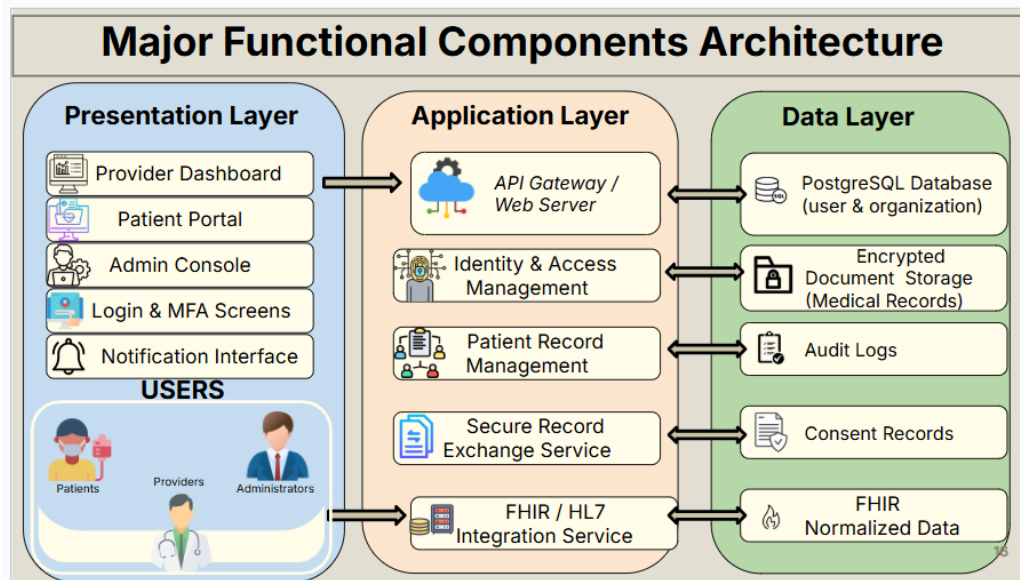
4.1 Prototype Architecture (Hardware/Software)

The PatientPath prototype will follow the same three-layer structure established for the real-world product: the Presentation Layer, Application Layer, and Data Layer. Separating the prototype into these layers allows the team to demonstrate how users interact with PatientPath, how the system processes those interactions, and how information is stored and retrieved.

The Presentation Layer contains the parts of PatientPath that users directly interact with. This includes the Provider Dashboard, Patient Portal, Admin Console, Login and MFA screens, and Notification Interface. The interface available to a user will depend on their assigned role. For example, providers will use the dashboard to create and manage referrals, while patients will use the Patient Portal to view referral information and manage consent.

The Application Layer handles the functionality behind those interfaces. This includes Secure Record Exchange, Identity and Access Management, Patient Record Management, FHIR/HL7 Integration, and the API Gateway. The Data Layer supports these functions by managing information through Encrypted Document Storage, Audit Logs, Consent Records, FHIR Normalized Data, and a PostgreSQL database.

The prototype will primarily be software-based and will not require specialized hardware. Users will be able to access PatientPath through a computer with an internet connection and web browser. The prototype MFCD will visually show how the Presentation, Application, and Data layers communicate with one another. It will also distinguish between components that are fully demonstrated and those that are simulated, reduced, or outside the prototype's scope.



4.2 Prototype Features and Capabilities

The PatientPath prototype will demonstrate the core features necessary to follow a patient referral through the system. Providers will be able to create referrals, select patient records to share, and view referral statuses as they move through stages such as Pending, In Review, Accepted, and Complete. Patients will be able to view relevant referral information and manage consent, while administrators will be able to monitor referral activity. The system will also identify referrals that have been inactive for too long so that they are less likely to be overlooked.

These capabilities are important because they demonstrate how PatientPath addresses communication and visibility problems within the referral process. Instead of relying on separate phone calls, faxes, and manual follow-ups, referral information can be organized within one system. Providers can see where a referral is in the process, while patients have greater visibility into their care and control over access to their information.

The success of the prototype will be demonstrated by completing the major PatientPath workflows. A provider should be able to create and track a referral, associate the necessary records with that referral, and see changes in referral status. Patients should be able to interact with their consent controls, and administrators should be able to monitor referral activity. Successfully completing these workflows will demonstrate that the major components of PatientPath can work together as one connected process.

The prototype also helps reduce risks associated with developing the real-world product. Instead of connecting to live healthcare systems or using actual patient information, these interactions can be modeled using test data. Security concepts can still be demonstrated through authentication, role-based access, consent management, and activity tracking without introducing protected health information. Interoperability concepts can also be demonstrated without requiring live connections to external EHR systems.

Table 1*Table of Comparison Between RWP and Prototype*

Feature	Prototype	Real World Product
Referral creation & tracking	Functional using test referral data and status updates	Uses live patient referrals between healthcare organizations
Provider dashboard	Functional dashboard for creating, viewing, and managing referrals	Full dashboard connected to live patient and referral information
Patient portal	Allows test users to view referral information and manage consent	Allows patients to view referral information and manage access to their health information
Admin console	Supports referral monitoring auditing, and role based access	Full administrative monitoring, auditing, and user management
Patient consent	Demonstrated using test users and simulated patient information	Manages real patient consent and access permissions
Role-based access control	Demonstrated for PCP, Specialist, Admin, and patient roles	Controls system access for verified users based on their assigned roles

Record sharing	Partially implemented or simulated	Integrates with external healthcare systems using FHIR and HL7
Multi-factor authentication	Partially implemented	Fully implemented for secure user authentication
Referral alerts	Identifies test referrals that have been inactive or require follow-up	Provides alerts based on live referral activity
Audit logs	Demonstrated using prototype user activity	Maintains activity history for security, compliance, and auditing
Database	PostgreSQL database containing test data	Production database containing secured patient and referral information
Encrypted document storage	Partially implemented	Securely stores real patient documents and records
Encrypted document storage	Partially implemented	Securely stores real patient documents and records
EHR integration	Simulated; no connection to live EHR systems	Connects patient path with participating healthcare organizations' EHR systems
Hosting & infrastructure	Prototype development environment	Multi-cloud, highly available production infrastructure

Table 1 shows the differences between the PatientPath prototype and the intended real-world product. The prototype focuses on demonstrating the main referral, record-sharing, consent, and monitoring workflows while reducing or simulating features that require access to real healthcare systems or patient information. These limitations allow the team to demonstrate PatientPath's core functionality within the scope of the project while still representing how the system could operate in a real healthcare environment.

4.3 Prototype Development Challenges

Several challenges are expected while developing the PatientPath prototype. One of the largest challenges is demonstrating healthcare interoperability without having direct access to real EHR systems. PatientPath is designed to work alongside existing healthcare systems using standards such as FHIR and HL7, but connecting the student prototype to actual healthcare organizations would require access, security, and resources that are outside the scope of the project. Because of this, some interoperability features will need to be simulated or partially implemented.

Security and privacy are another challenge because PatientPath is designed around sensitive healthcare information. The prototype cannot use real patient health information, so test data will be used to demonstrate record sharing, consent, and referral workflows. Features such as authentication, role-based access, consent controls, and activity tracking will be used to demonstrate how security would function without creating the risks associated with storing actual patient data.

Another challenge will be connecting the different parts of the prototype as development continues. The Provider Dashboard, Patient Portal, Admin Console, database, and backend functionality must work together for the referral process to feel like one connected system. Since different team members are responsible for different components, maintaining consistent interfaces and making sure each component works correctly with the others will require communication, testing, and integration throughout development.

Finally, the team must balance the desired functionality of PatientPath with the time and technical limitations of a semester-long project. Some production-level features will therefore remain simulated, reduced, or outside the scope of the prototype. The team's priority will be making sure the major referral, record-sharing, consent, and monitoring workflows function well enough to demonstrate PatientPath's overall purpose.

5 Glossary

- **Application Programming Interface (API):** a set of protocols and tools that allows different software applications to communicate and exchange data.
- **Continuous Integration / Continuous Deployment (CI/CD):** a software development practice that automates the integration, testing, and deployment of code changes.
- **Electronic Health Record (EHR):** a digital record of a patient's medical and health information.
- **Fast Healthcare Interoperability Resources (FHIR):** an HL7 standard designed to enable the electronic exchange of healthcare information.
- **Federally Qualified Health Center (FQHC):** a healthcare organization that provides primary care services to underserved communities and meets federal requirements.
- **Health Information Exchange (HIE):** the electronic sharing of health-related information among healthcare organizations and providers.
- **Health Insurance Portability and Accountability Act (HIPAA):** a federal law that establishes standards for protecting the privacy and security of individuals' health information.
- **Health Level Seven (HL7):** an international healthcare standards organization that develops standards for the exchange and integration of electronic health information.
- **Multi-Factor Authentication (MFA):** a security method that requires users to provide two or more forms of verification before gaining access to a system.
- **Primary Care Provider (PCP):** a healthcare professional who provides general medical care and serves as a patient's primary point of contact for healthcare services.
- **Representational State Transfer (REST):** an architectural style commonly used to design web-based APIs and facilitate communication between systems.
- **Role-Based Access Control (RBAC):** a security approach that restricts system access based on a user's assigned role and associated permissions.

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